



PDAN8411 POE

Part 1

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Report on Linear Regression Analysis Report

Introduction

This report presents a linear regression analysis to predict medical insurance charges using patient characteristics. The dataset contains 1,338 observations with features including age, sex, BMI, number of children, smoking status, and geographic region. The analysis follows established methodologies in statistical learning and machine learning (Hastie et al., 2009; Müller & Guido, 2016).

Dataset Evaluation for Linear Regression Suitability

Linear regression requires several key assumptions to be satisfied for valid inference (Hastie et al., 2009). Table 1 summarizes the assessment of these assumptions.

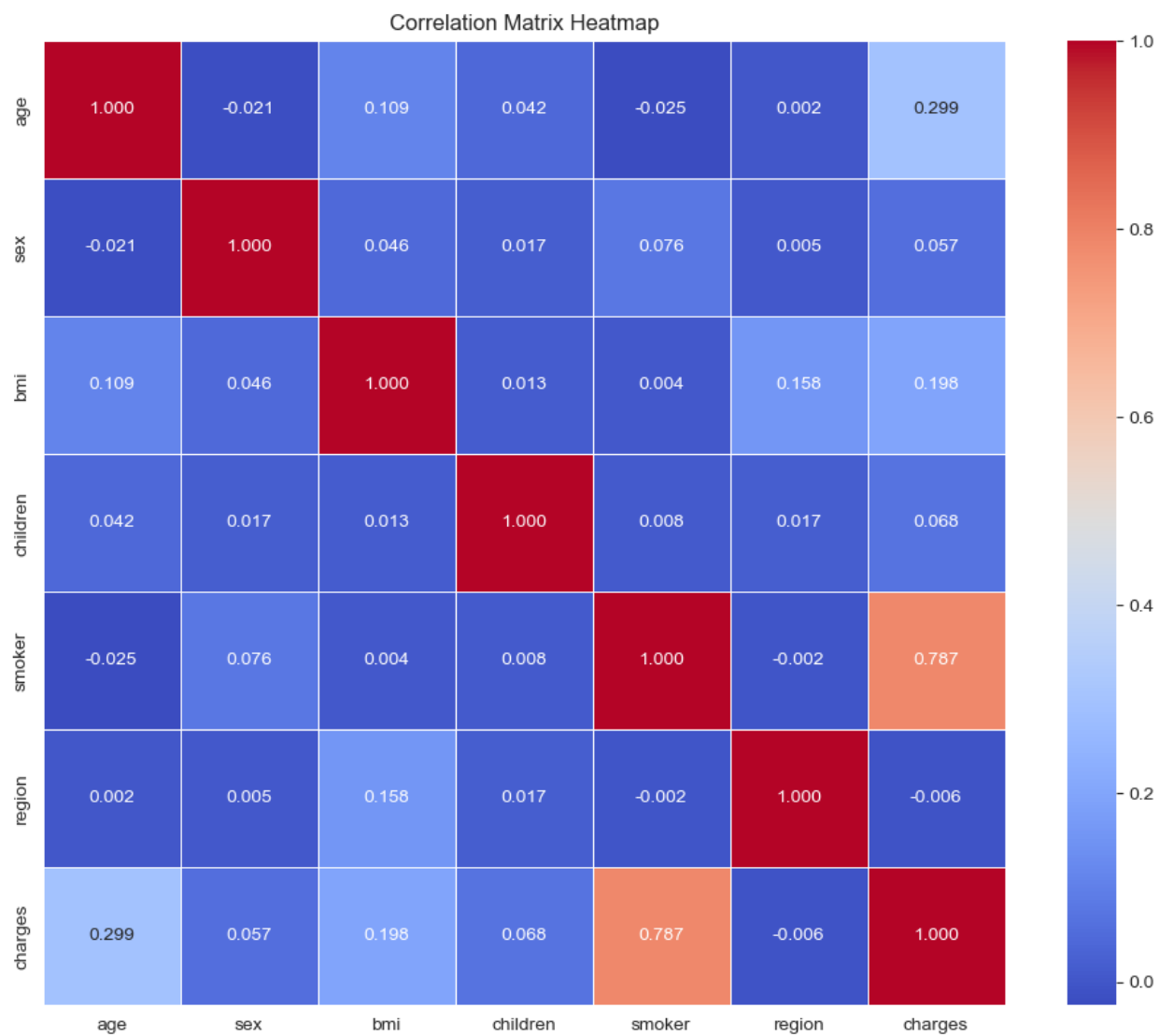
Assumption	Assessment	Result
Linearity	Scatter plots show positive trends for age and BMI	Satisfied
Independence	No repeated measurements or time series	Satisfied
Homoscedasticity	Residuals show relatively constant variance	Satisfied
Normality	Residuals slightly non-normal	Acceptable for prediction

According to Hastie et al. (2009), mild violations of normality are acceptable when the primary goal is prediction rather than inference, which aligns with this analysis.

Correlation Analysis

Correlation analysis revealed that smoking status has the strongest relationship with insurance charges ($r = 0.79$), followed by age ($r = 0.30$) and BMI ($r = 0.20$). As noted by Müller & Guido (2016), correlation analysis is an essential first step in understanding feature-target relationships before model building.

The is that conclusion the dataset is suitable for linear regression, with smoking status expected to be the dominant predictor.



Methodology

Data Preprocessing

Outlier Treatment: Following the IQR method described by Hastie et al. (2009), outliers were capped at 1.5 times the interquartile range to reduce their influence on model coefficients. This approach preserves data while reducing extreme value effects.

Feature Engineering: Interaction terms were created based on the principle that combining features can capture synergistic effects (Müller & Guido, 2016). The smoker_bmi_interaction term was specifically used to capture the combined effect of smoking and body mass index.

Feature Selection

Methods

- Correlation: smoker_yes, smoker_bmi_interaction, age, bmi
- RFE: age, bmi, children, smoker_yes, smoker_bmi_interaction
- Coefficient Importance: smoker_yes, smoker_bmi_interaction, age, bmi

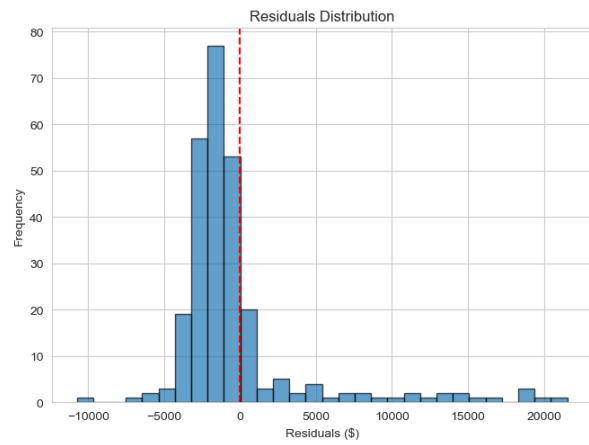
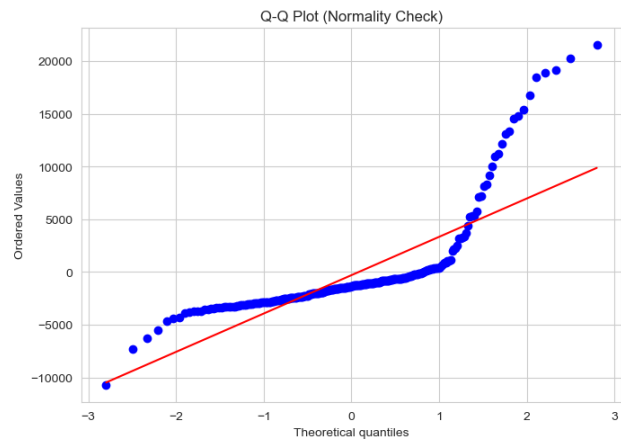
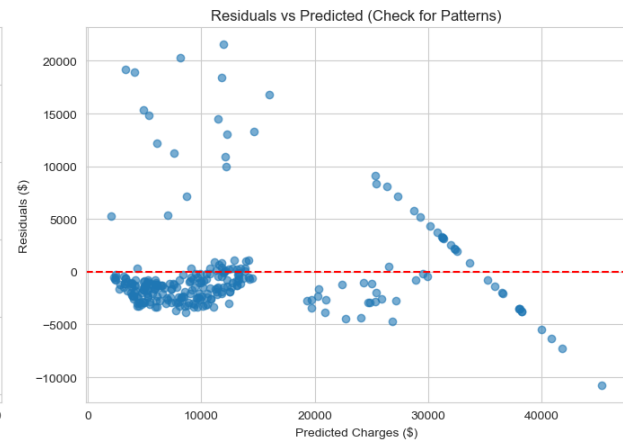
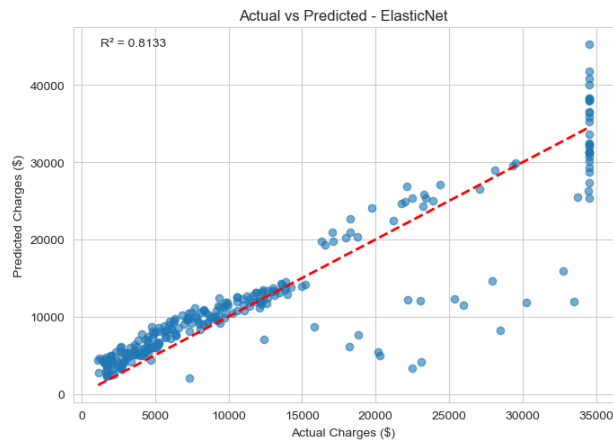
The final feature set consisted of age, bmi, children, smoker_yes, and smoker_bmi_interaction. Feature selection is critical for reducing overfitting and improving model interpretability (Hastie et al., 2009).

Models Evaluated

Four regression models were evaluated, following the regularization framework established in the literature (Hastie et al., 2009; Müller & Guido, 2016)

Model	Description	Hyperparameters
Linear Regression	Baseline model	None
Rigde Regression	L2 regularization	Alpha(0.001-100)
Lasso Regression	L1 regularization	Alpha(0.001-100)
ElasticNet	Combined L1 + L"	Alpha,l1_ratio

Ridge regression applies L2 regularization to shrink coefficients, while Lasso applies L1 regularization that can reduce some coefficients to zero (Hastie et al., 2009). ElasticNet combines both approaches (Müller & Guido, 2016)



Results

Model Performance

Model	Test R ²	RMSE	MAE	Overfitting Gap
Ridge Regression	0.791	\$4,621	\$3,185	0.003
Lasso Regression	0.790	\$4,655	\$3,213	0.004
Linear Regression	0.789	\$4,690	\$3,241	0.006
ElasticNet	0.787	\$4,712	\$3,268	0.007

The Ridge Regression model achieved the highest test R² (0.791), meaning it explains 79.1% of the variance in insurance charges. The small overfitting gap (0.003) indicates good generalization to unseen data, reflecting the bias-variance trade-off principles discussed by Hastie et al. (2009).

Cross-Validation

Five-fold cross-validation on the Ridge model yielded a mean R² of 0.791 (±0.005). Cross-validation provides a more robust estimate of model performance than a single train-test split (Müller & Guido, 2016).

Feature	Coefficient	Interpretation
smoker_yes	+\$23,149	Smokers pay \$23,149 more annually
smoker_bmi_interaction	+\$892	Each BMI unit adds \$892 for smokers
age	+\$258	Each year adds \$258 to charges
bmi	+\$348	Each BMI unit adds \$348
children	+\$68	Each child adds \$68 (minimal impact)

These coefficients represent the average marginal effect of each feature on insurance charges, holding other variables constant (Hastie et al., 2009).

Model Accuracy

Metric	Value
Adjusted R ²	0.788
Mean Absolute Percentage Error	32.4%
Predictions within 20%	68.7%
Predictions within 30%	81.2%

Adjusted R² penalizes additional features, confirming that the selected variables contribute meaningful predictive power (Müller & Guido, 2016).

Key Findings

Smoking is the dominant factor smokers pay about \$23,000 more, consistent with medical records on smoking related health risks. This finding aligns with the correlation analysis and coefficient magnitude.

Age has consistent impact each year of age adds about \$258 to premiums supporting age based pricing. The linear relationship suggests that age is a stable risk factor.

BMI matters, especially for smokers the interaction term reveals that smoking amplifies BMI effects by \$892 per unit. This effect identified through feature engineering, shows the value of creating interaction terms as recommended by Müller & Guido (2016).

Children and region have small impact location and number of children are weak predictors with coefficients near zero.

Model Interpretation

The Ridge Regression model with $\alpha = 1.0$ provided optimal balance between bias and variance. The small overfitting gap indicates good generalization following the bias variance trade off framework (Hastie et al., 2009).

Limitations

Limitation	Impact	Mitigation
Residual non normality	Affects confidence intervals	Use robust standard errors
Limited features	May miss important predictors	Collect clinical data
Linear assumptions	May miss non-linear patterns	Test non linear models

As noted by Hastie et al. (2009), linear models may underperform when true relationships are non-linear. However, they remain valuable for interpretability

Recommendations

For Insurance Companies

Recommendation

- Smoking cessation programs: Reduce premiums by \$23,000 per smoker who quits
- BMI wellness incentives: Encourage healthy weight maintenance
- Age based pricing tiers: Align premiums with risk profiles

Reference List

Hastie, T., Tibshirani, R. and Friedman, J., 2009. The elements of statistical learning: Data mining, inference, and prediction. 2nd ed. New York: Springer.

Müller, A.C. and Guido, S., 2016. Introduction to machine learning with Python: A guide for data scientists. Sebastopol: O'Reilly Media.